EEG Singal Classification in

Neurological Disorders

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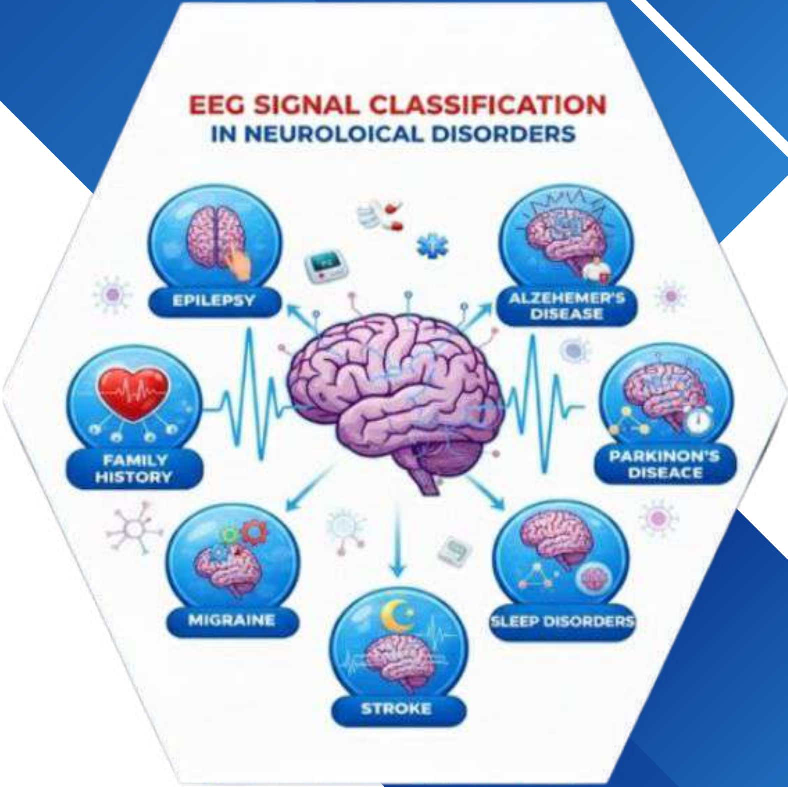

OUTLINE

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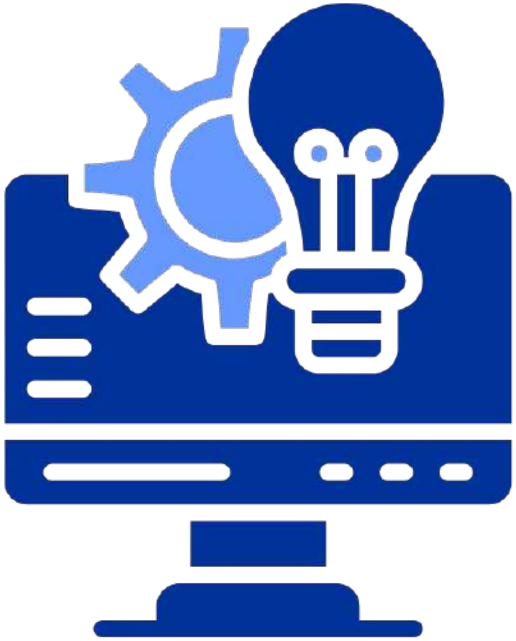


• Abstract:

EEG signals are used to analyze brain activity and diagnose neurological and mental health disorders Manual interpretation of EEG data is complex, time consuming and prone to human error.

The proposed system uses a Deep Learning- based LSTM model for automatic multi- class classification.

It detects multiple disorders including Anxiety, OCD, Addictive disorders and healthy conditions

EEG signals are preprocessed using filtering , normalization and segmentation to improve accuracy.

An LLM based explanation module provides human readable interpretations of predictions

Problem Statement :

EEG data is highly complex and difficult to interpret manually .
Manual diagnosis depends heavily on expert neurologists.
Large volumes of EEG data increase analysis time and effort .
Human errors may occur during manual interpretation.
Multiple neurological disorders show overlapping EEG patterns.
Existing automated systems lack explainability and transparency

Proposed Solution

Develop an automated EEG classification system using a Deep Learning –based LSTM model. Preprocess raw EEG signals through filtering, normalization and segmentation to improve data quality Train the LSTM model to perform accurate multi-class classification of neurological disorders. Integrate an LLM-based module to generate clear and huma –readable explanations for predictions. Implement a private blockchain using Ganache to ensure and tamper- proof result storage. Design a web-based client –server application for real-time EEG data upload prediction visualization

System Architecture

User Interface Layer

A web-based interface built using HTML and CSS allows users to upload EEG data, view predictions, and access explanations in real time.

Backend Application Layer

Developed using Flask, this layer handles file uploads, preprocessing, model execution, explanation generation, and communication with the blockchain.

AI & Processing Layer:

EEG signals are preprocessed and fed into the LSTM model for multi-class classification, while an LLM module generates prediction explanations.

Blockchain & Storage Layer:

Classification results are securely stored using a private Ganache blockchain to ensure data integrity and tamper-proof record keeping.

Live Demo of Project

Here's the list of some demo of the projects -

Code

I recently explored a **heart disease risk prediction dataset** to understand the key factors affecting cardiovascular health. The goal was to identify patterns and insights that can help in **early detection and prevention**.

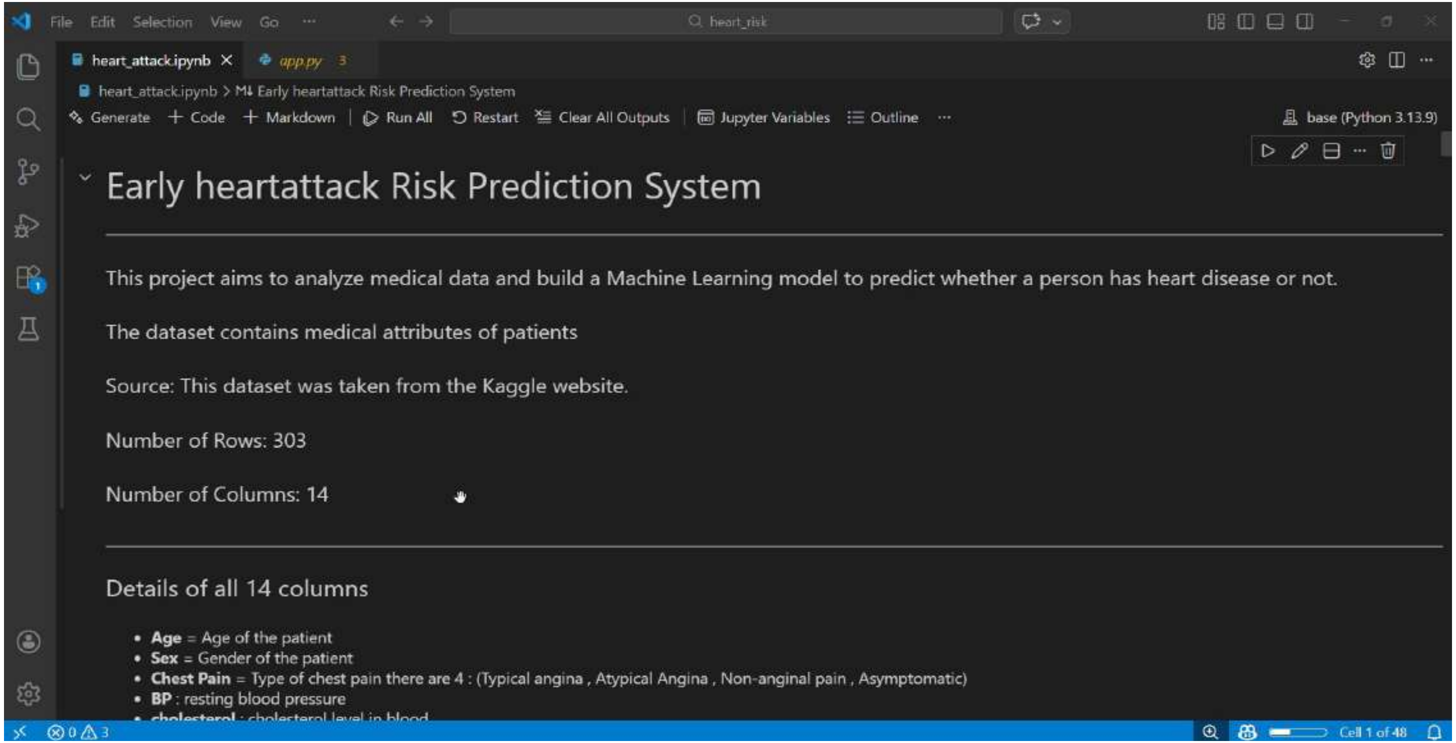
Input

The inputs we have chosen for this dataset are Age, cholesterol , BP , chest pain……

Output

The output is to predict the risk percentage of the heart attack based on the inputs and visualization

Video of Project Demo



Conclusion

The proposed EEG signal classification system successful automates the detection of multiple neurological and mental health disorders using an LSTM-based deep learning model. The system reduces manual effort, improves diagnostic accuracy, and efficiently handles complex EEG time-series data. By integrating an LLM-based explanation module, it enhances transparency and clinical trust in predictions. The use of a private blockchain ensures secure, tamper-proof storage of classification results, maintaining data integrity. Overall, the project combines deep learning, explainable AI, and blockchain technology to provide a reliable, secure, and intelligent framework for automated neurological disorder detection.

Future Scope

The proposed EEG classification system can be enhanced to support real-time monitoring and live disorder detection. The model can be expanded to identify additional neurological and psychiatric conditions with improved accuracy. Integration with hospital systems and cloud platforms can increase scalability and accessibility. Future improvements may also include advanced AI-based predictive analysis, mobile application support, and IoT-enabled EEG device integration for remote healthcare use.

Reference:

https://www.kaggle.com/datasets/neurocipher/heartdisease -
Kaggle

https://github.com/Deepika-Ramala/Student-project/upload
Git Hub

THANK YOU